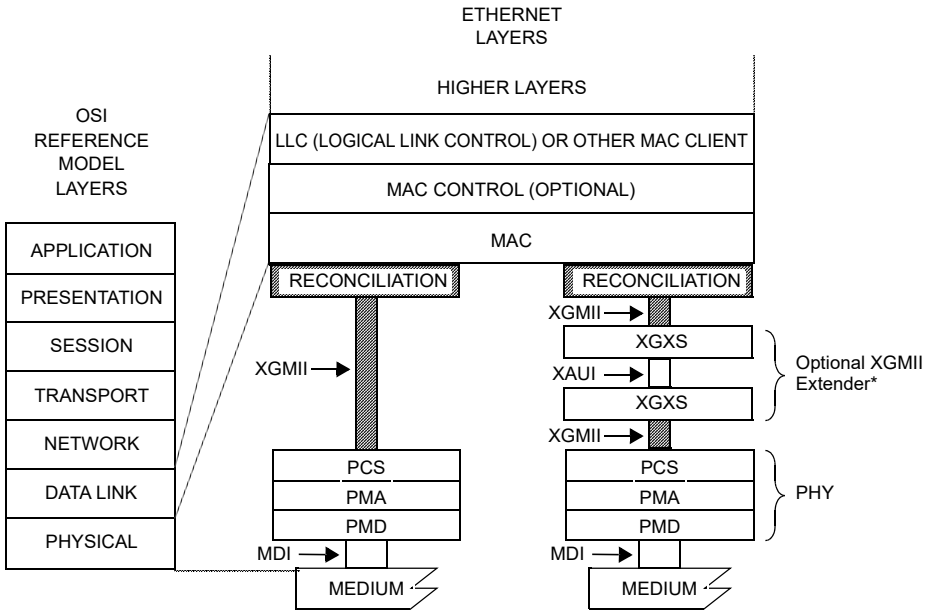
46. Reconciliation Sublayer (RS) and 10 Gigabit Media Independent Interface (XGMII)

46.1 Overview

This clause defines the logical and electrical characteristics for the Reconciliation Sublayer (RS) and 10 Gigabit Media Independent Interface (XGMII) between Ethernet media access controllers and various PHYs. Figure 46–1 shows the relationship of the RS and XGMII to the ISO/IEC (IEEE) OSI reference model.



MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
XAUI = 10 GIGABIT ATTACHMENT UNIT INTERFACE
XGMII = 10 GIGABIT MEDIA INDEPENDENT INTERFACE
XGXS = XGMII EXTENDER SUBLAYER

*specified in Clause 47

Figure 46–1—XGMII relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

The purpose of the XGMII is to provide a simple, inexpensive, and easy-to-implement interconnection between the Media Access Control (MAC) sublayer and the Physical Layer device (PHY). The 10 Gigabit Attachment Unit Interface (XAUI) may optionally be used to extend the operational distance of the XGMII with reduced pin count (see Clause 47).

The RS adapts the bit serial protocols of the MAC to the parallel encodings of 2.5 Gb/s, 5 Gb/s, and 10 Gb/s PHYs. Though the XGMII is an optional interface, it is used extensively in this standard as a basis for specification. The 2.5 Gb/s, 5 Gb/s, and 10 Gb/s Physical Coding Sublayers (PCS) are specified to the XGMII, so if not implemented, a conforming implementation shall behave functionally as if the RS and XGMII were implemented.

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The XGMII has the following characteristics:

a) It is capable of supporting at least one of the following rates of operation: 2.5 Gb/s, 5 Gb/s, or 10 Gb/s.
b) Data and delimiters are synchronous to clock reference.
c) It provides independent 32-bit-wide transmit and receive data paths.
d) It uses signal levels compatible with common digital ASIC processes.
e) It provides for full duplex operation only.

46.1.1 Summary of major concepts

The following are the major concepts of XGMII:

a) The XGMII is functionally similar to the MII defined in Clause 22 and GMII defined in Clause 35 as they all define an interface allowing independent development of MAC and PHY logic.
b) The RS converts between the MAC serial data stream and the parallel data paths of the XGMII.
c) The RS maps the signal set provided at the XGMII to the PLS service primitives provided at the MAC.
d) Each direction of data transfer is independent and serviced by data, control, and clock signals.
e) The RS generates continuous data or control characters on the transmit path and expects continuous data or control characters on the receive path.
f) The RS participates in link fault detection and reporting by monitoring the receive path for status reports that indicate an unreliable link, and generating status reports on the transmit path to report detected link faults to the DTE on the remote end of the connecting link.
g) When the XGMII is optionally extended with XAUI, two XGMII interfaces logically exist (see Figure 46–1). The transmit path signals are from the RS to the DTE (top) XGXS of the XAUI via one XGMII and from the PHY (bottom) XGXS to the PCS via the other XGMII. The receive path signals are from the PCS to the PHY XGXS of the XAUI via one XGMII and from the DTE XGXS to the RS via the other XGMII. The descriptions of the XGMII as between the RS and the PCS are, therefore, equally applicable between the RS and the DTE XGXS or the PHY XGXS and the PCS.
h) The XGMII may also support Low Power Idle (LPI) signaling for PHY types supporting Energy-Efficient Ethernet (EEE) (see Clause 78).
i) The XGMII is rate scalable and may support rates of 2.5 Gb/s, 5 Gb/s, and 10 Gb/s.

46.1.2 Application

This clause applies to the interface between the MAC and PHY. The physical implementation of the interface is primarily intended as a chip-to-chip (integrated circuit to integrated circuit) interface implemented with traces on a printed circuit board. The XGMII may also be used in other ways, for example, as a logical interface between ASIC logic modules within an integrated circuit.

This interface is used to provide media independence so that an identical media access controller may be used with all 2.5GBASE, 5GBASE, and 10GBASE PHY types.

46.1.3 Rate of operation

The XGMII supports MAC data rates of 2.5 Gb/s, 5 Gb/s, and 10 Gb/s as defined within this clause. A compliant device may implement any subset of these rates. Operation at 10 Mb/s and 100 Mb/s is supported by the MII defined in Clause 22 and operation at 1000 Mb/s by the GMII defined in Clause 35.

10GBASE-X and 10GBASE-R PHYs operate at a 10 Gb/s data rate. 10GBASE-W PHYs operate at the STS-192/VC-4-64c line rate of 9.95328 Gb/s, mapping the encoded data stream at a 9.58464 Gb/s payload rate. On transmit, this mapping is performed by discarding Idle control characters corresponding to the

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stretched interpacket gap created by the MAC in this mode of operation, and on receive, by adding interpacket gap Idle control characters as required to adapt to the XGMII RX_CLK rate.

46.1.4 Delay constraints

Predictable operation of the MAC Control PAUSE operation (Clause 31, Annex 31B) demands that there be an upper bound on the propagation delays through the network. This implies that MAC, MAC Control sublayer, and PHY implementations conform to certain delay maxima, and that network planners and administrators conform to constraints regarding the cable topology and concatenation of devices. The maximum cumulative MAC Control, MAC and RS round-trip (sum of transmit and receive) delay shall meet the values specified in Table 46–1. Bit time is specified in 1.4, and pause_quanta is specified in 31B.2.

Table 46–1—Round-trip delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)
MAC, RS, and MAC Control	8192	16

46.1.5 Allocation of functions

The allocation of functions at the XGMII balances the need for media independence with the need for a simple and cost-effective interface. The bus width and signaling rate are applicable to short distance chip-to-chip interconnect with printed circuit board trace lengths electrically limited to approximately 7 cm. The XGMII (like the MII and GMII) maximizes media independence by cleanly separating the Data Link and Physical Layers of the ISO (IEEE) seven-layer reference model. This allocation also recognizes that implementations can benefit from a close coupling between the PLS sublayer or PCS and the PMA sublayer.

46.1.6 XGMII structure

The XGMII is composed of independent transmit and receive paths. Each direction uses 32 data signals (TXD<31:0> and RXD<31:0>), four control signals (TXC<3:0> and RXC<3:0>), and a clock (TX_CLK and RX_CLK). Figure 46–2 depicts a schematic view of the RS inputs and outputs.

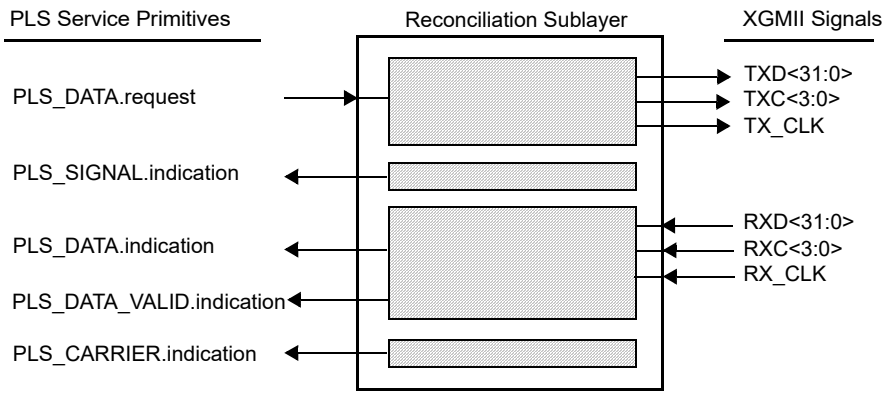


Figure 46–2—Reconciliation Sublayer (RS) inputs and outputs

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The 32 TXD and four TXC signals shall be organized into four data lanes, as shall the 32 RXD and four RXC signals (see Table 46–2). The four lanes in each direction share a common clock—TX_CLK for transmit and RX_CLK for receive. The four lanes are used in round-robin sequence to carry an octet stream. On transmit, each eight PLS_DATA.request transactions represent an octet transmitted by the MAC. The first octet is aligned to lane 0, the second to lane 1, the third to lane 2 the fourth to lane 3, then repeating with the fifth to lane 0, etc. Delimiters and interframe idle characters are encoded on the TXD and RXD signals with the control code indicated by assertion of TXC and RXC, respectively.

Table 46–2—Transmit and receive lane associations

TXD RXD	TXC RXC	Lane
<7:0>	<0>	0
<15:8>	<1>	1
<23:16>	<2>	2
<31:24>	<3>	3

46.1.7 Mapping of XGMII signals to PLS service primitives

The Reconciliation Sublayer (RS) shall map the signals provided at the XGMII to the PLS service primitives defined in Clause 6. The PLS service primitives provided by the RS and described here behave in exactly the same manner as defined in Clause 6. Full duplex operation only is implemented at 2.5 Gb/s, 5 Gb/s, and 10 Gb/s; therefore, PLS service primitives supporting CSMA/CD operation are not mapped through the RS to the XGMII. The mapping is changed if EEE capability is supported (see 78.3). This behavior and restrictions are the same as described in 22.7, with the details of the signaling described in 46.3. LPI_REQUEST shall not be set to ASSERT unless the attached link has been operational for at least one second (i.e., link_status = OK, according to the underlying PCS/PMA).

EEE capability requires the use of the MAC defined in Annex 4A for simplified full duplex operation (with carrier sense deferral). This provides full duplex operation but uses the carrier sense signal to defer transmission when the PHY is in its low power state.

Mappings for the following primitives are defined for 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation:

PLS_DATA.request
PLS_DATA.indication
PLS_CARRIER.indication
PLS_SIGNAL.indication
PLS_DATA_VALID.indication

46.1.7.1 Mapping of PLS_DATA.request

46.1.7.1.1 Function

Map the primitive PLS_DATA.request to the XGMII signals TXD<31:0>, TXC<3:0>, and TX_CLK.

46.1.7.1.2 Semantics of the service primitive

PLS_DATA.request (OUTPUT_UNIT)

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The OUTPUT_UNIT parameter can take one of three values: ONE, ZERO, or DATA_COMPLETE. It represents a single data bit. The DATA_COMPLETE value signifies that the Media Access Control sublayer has no more data to output.

46.1.7.1.3 When generated

This primitive is generated by the MAC sublayer to request the transmission of a single data bit on the physical medium or to stop transmission.

46.1.7.1.4 Effect of receipt

The OUTPUT_UNIT values are conveyed to the PHY by the signals TXD<31:0> and TXC<3:0> on each TX_CLK edge. Each PLS_DATA.request transaction shall be mapped to a TXD signal in sequence (TXD<0>, TXD<1>,... TXD<31>, TXD<0>) as described in 46.2. After 32 PLS_DATA.request transactions from the MAC sublayer (four octets of eight PLS_DATA.request transactions each), the RS requests transmission of 32 data bits by the PHY. The first octet of preamble shall be converted to a Start control character and aligned to lane 0. The TXD<31:0> and TXC<3:0> shall be generated by the RS for each 32 bit-times of the MAC sublayer.

The DATA_COMPLETE value shall be mapped to a Terminate control character encoded on the next eight TXD signals in sequence after the last data octet; and is transferred to the PHY at the next TX_CLK edge. This may be on the same TX_CLK edge as the last data octet or the subsequent TX_CLK edge. When the Terminate control character is in lane 0, 1, or 2, the lanes following in sequence are encoded with an Idle control character.

46.1.7.2 Mapping of PLS_DATA.indication

46.1.7.2.1 Function

Map the primitive PLS_DATA.indication to the XGMII signals RXD<31:0>, RXC<3:0> and RX_CLK.

46.1.7.2.2 Semantics of the service primitive

PLS_DATA.indication (INPUT_UNIT)

The INPUT_UNIT parameter can take one of two values: ONE or ZERO. It represents a single data bit.

46.1.7.2.3 When generated

The INPUT_UNIT values are derived from the signals RXC<3:0> and RXD<31:0> received from the PHY on each edge of RX_CLK. Each primitive generated to the MAC sublayer entity corresponds to a PLS_DATA.request issued by the MAC at the remote end of the link connecting two DTEs. For each RXD<31:0> during frame reception, the RS shall generate 32 PLS_DATA.indication transactions until the end of frame (Terminate control character), where 0, 8, 16, or 24 PLS_DATA.indication transactions will be generated from the RXD<31:0> containing the Terminate. During frame reception, each RXD signal shall be mapped in sequence into a PLS_DATA.indication transaction (RXD<0>, RXD<1>,... RXD<31>, RXD<0>) as described in 46.2.

The RS shall convert a valid Start control character to a preamble octet prior to generation of the associated PLS_DATA.indication transactions. The RS shall not generate any PLS_DATA.indication primitives for a Terminate control character. To assure robust operation, the value of the data transferred to the MAC may be changed by the RS as required by XGMII error indications (see 46.3.3). Sequence ordered sets are not indicated to the MAC (see 46.3.4).

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46.1.7.2.4 Effect of receipt

The effect of receipt of this primitive by the MAC sublayer is unspecified.

46.1.7.3 Mapping of PLS_CARRIER.indication

2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation supports full duplex operation only. The RS never generates this primitive for PHYs that do not support EEE or Link Interruption.

For PHYs that support EEE capability, CARRIER_STATUS is set in response to LPI_REQUEST as shown in Figure 46–13. For PHYs that support Link Interruption, CARRIER_STATUS may be set in response to link_fault. CARRIER_STATUS is set to CARRIER_ON if LPI_CARRIER_STATUS is TRUE or if link_fault is Link Interruption. CARRIER_STATUS is otherwise set to CARRIER_OFF. The deferral mechanism based upon the Link Interruption signal may be enabled or disabled by management.

46.1.7.4 Mapping of PLS_SIGNAL.indication

2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation supports full duplex operation only. The RS never generates this primitive.

46.1.7.5 Mapping of PLS_DATA_VALID.indication

46.1.7.5.1 Function

Map the primitive PLS_DATA_VALID.indication to the XGMII signals RXC<3:0> and RXD<31:0>.

46.1.7.5.2 Semantics of the service primitive

PLS_DATA_VALID.indication (DATA_VALID_STATUS)

The DATA_VALID_STATUS parameter can take one of two values: DATA_VALID or DATA_NOT_VALID. The DATA_VALID value indicates that the INPUT_UNIT parameter of the PLS_DATA.indication primitive contains a valid data of an incoming frame. The DATA_NOT_VALID value indicates that the INPUT_UNIT parameter of the PLS_DATA.indication primitive does not contain valid data of an incoming frame.

46.1.7.5.3 When generated

The PLS_DATA_VALID.indication service primitive shall be generated by the RS whenever the DATA_VALID_STATUS parameter changes from DATA_VALID to DATA_NOT_VALID or vice versa.

DATA_VALID_STATUS shall assume the value DATA_VALID when a PLS_DATA.indication transaction is generated in response to reception of a Start control character on lane 0 if the prior RXC<3:0> and RXD<31:0> contained four Idle characters or a Sequence ordered set. DATA_VALID_STATUS shall assume the value DATA_NOT_VALID when RXC of the current lane in sequence is asserted for anything except an Error control character. In the absence of errors, DATA_NOT_VALID is caused by a Terminate control character. When DATA_VALID_STATUS changes from DATA_VALID to DATA_NOT_VALID because of a control character other than Terminate, the RS shall ensure that the MAC will detect a FrameCheckError prior to indicating DATA_NOT_VALID to the MAC (see 46.3.3.1).

46.1.7.5.4 Effect of receipt

The effect of receipt of this primitive by the MAC sublayer is unspecified.

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46.2 XGMII data stream

Packets transmitted through the XGMII shall be transferred within the XGMII data stream. The data stream is a sequence of bytes, where each byte conveys either a data octet or control character. The parts of the data stream are shown in Figure 46–3.

<inter-frame><preamble><sfd><data><efd>

Figure 46–3—XGMII data stream

For the XGMII, transmission and reception of each bit and mapping of data octets to lanes shall be as shown in Figure 46–4.

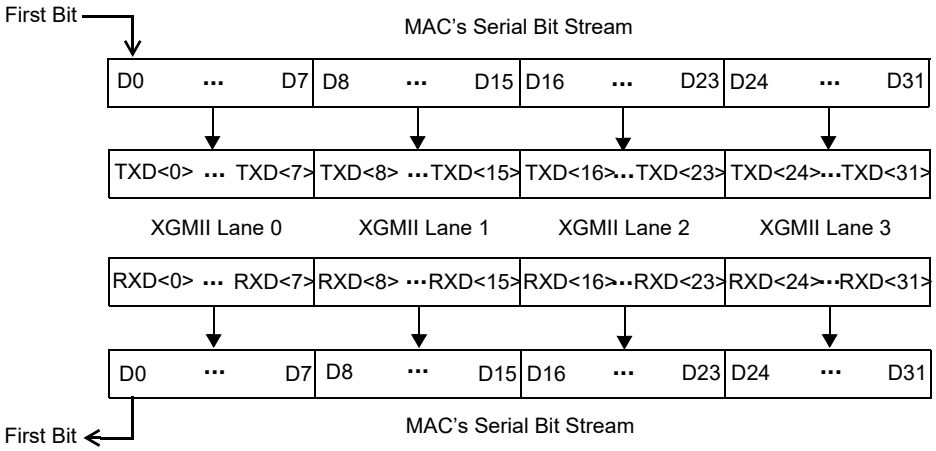


Figure 46–4—Relationship of data lanes to MAC serial bit stream

46.2.1 Inter-frame <inter-frame>

The inter-frame <inter-frame> period on an XGMII transmit or receive path is an interval during which no frame data activity occurs. The <inter-frame> corresponding to the MAC interpacket gap begins with the Terminate control character, continues with Idle control characters and ends with the Idle control character prior to a Start control character. The length of the interpacket gap may be changed between the transmitting MAC and receiving MAC by one or more functions (e.g., RS lane alignment, PHY clock rate compensation, or 10GBASE-W data rate adaptation functions). The minimum IPG at the XGMII of the receiving RS is five octets.

The signaling of link status information logically occurs in the <inter-frame> period (see 46.3.4). Subclause 46.3.3 describes frame processing when signaling of link status information is initiated or terminated.

46.2.2 Preamble <preamble> and start of frame delimiter <sfd>

The preamble <preamble> begins a frame transmission by a MAC as specified in 4.2.5 and when generated by a MAC consists of 7 octets with the following bit values:

10101010 10101010 10101010 10101010 10101010 10101010 10101010

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The Start control character indicates the beginning of MAC data on the XGMII. On transmit, the RS converts the first data octet of preamble transferred from the MAC into a Start control character. On receive, the RS will convert the Start control character into a preamble data octet. The start control character is aligned to lane 0 of the XGMII by the RS on transmit and by the PHY on receive.

The start of frame delimiter <sfd> indicates the start of a frame and immediately follows the preamble. The bit value of <sfd> at the XGMII is unchanged from the Start Frame Delimiter (SFD) specified in 4.2.6 and is the bit sequence:

10101011

The preamble and SFD are shown previously with their bits ordered for serial transmission from left to right. As shown, the left-most bit of each octet is the LSB of the octet and the right-most bit of each octet is the MSB of the octet.

The preamble and SFD are transmitted through the XGMII as octets sequentially ordered on the lanes of the XGMII. The first preamble octet is replaced with a Start control character and it is aligned to lane 0, the second octet on lane 1, the third on lane 2 and the fourth on lane 3, and the four octets are transferred on the next edge of TX_CLK. The fifth octet is assigned to lane 0 with subsequent octets sequentially assigned to the lanes with the SFD assigned to lane 3. The XGMII <preamble> and <sfd> are as follows:

Lane 0	**Lane 1**	**Lane 2**	**Lane 3**
Start	10101010	10101010	10101010
10101010	10101010	10101010	10101011

46.2.3 Data <data>

The data <data> in a well-formed frame shall consist of a set of data octets.

46.2.4 End of frame delimiter <efd>

Assertion of TXC with the appropriate Terminate control character encoding of TXD on a lane constitutes an end of frame delimiter <efd> for the transmit data stream. Similarly, assertion of RXC with the appropriate Terminate control character encoding of RXD constitutes an end of frame delimiter for the receive data stream. The XGMII shall recognize the end of frame delimiter on any of the four lanes of the XGMII.

46.2.5 Definition of Start of Packet and End of Packet Delimiters

For the purposes of Clause 30, the Start of Packet delimiter is defined as the Start control character, and the End of Packet delimiter is defined as the end of the last sequential data octet preceding the Terminate control character or other control character causing a change from DATA_VALID to DATA_NOT_VALID. (See 46.1.7.5.2 and 30.3.2.1.5.)

46.3 XGMII functional specifications

The XGMII is designed to make the differences among the various media and transceiver combinations transparent to the MAC sublayer. The selection of logical control signals and the functional procedures are all designed to this end.

NOTE—No XGMII loopback is defined, but XGMII signals are specified such that transmit signals may be connected to receive signals to create a loopback path. To do this, TXD<0> is connected to RXD<0> ... TXD<31> to RXD<31>, TXC<0> to RXC<0> ... TXC<3> to RXC<3>, and TXCLK to RXCLK. Such a loopback does not test the Link Fault Signaling state diagram, nor any of the error handling functions of the receive RS.

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46.3.1 Transmit

46.3.1.1 TX_CLK (transmit clock)

TX_CLK is a continuous clock. TX_CLK provides the timing reference for the transfer of the TXC<3:0> and TXD<31:0> signals from the RS to the PHY. The values of TXC<3:0> and TXD<31:0> shall be sampled by the PHY on both the rising edge and falling edge of TX_CLK. TX_CLK is sourced by the RS.

The TX_CLK frequency shall be $1/64 \times f_{MAC} \pm 100$ ppm, where f_{MAC} is the frequency (in Hz) corresponding to the MAC's nominal bit rate.

NOTE—For EEE capability, TX_CLK may be halted according to 46.3.1.5.

46.3.1.2 TXC<3:0> (transmit control)

TXC<3:0> indicate that the RS is presenting either data or control characters on the XGMII for transmission. The TXC signal for a lane shall be de-asserted when a data octet is being sent on the corresponding lane and asserted when a control character is being sent. In the absence of errors, the TXC signals are de-asserted by the RS for each octet of the preamble (except the first octet that is replaced with a Start control character) and remain de-asserted while all octets to be transmitted are presented on the lanes of the XGMII. TXC<3:0> are driven by the RS and shall transition synchronously with respect to both the rising and falling edges of TX_CLK. Table 46–3 specifies the permissible encodings of TXD and TXC for a XGMII transmit lane. Additional requirements apply for proper code sequences and in which lanes particular codes are valid (e.g., Start control character is to be aligned to lane 0).

A PHY with EEE capability shall interpret the combination of TXC and TXD as shown in Table 46–3 as an assertion of LPI. Transition into and out of the LPI state is shown in Figure 46–7.

46.3.1.3 TXD<31:0> (transmit data)

TXD is a bundle of 32 data signals organized into four lanes of eight signals each (TXD<7:0>, TXD<15:8>, TXD<23:16>, and TXD<31:24>) that are driven by the RS. Each lane is associated with a TXC signal as shown in Table 46–2 and shall be encoded as shown in Table 46–3. TXD<31:0> shall transition synchronously with respect to both the rising and falling edges of TX_CLK. For each high or low TX_CLK transition, data and/or control are presented on TXD<31:0> to the PHY for transmission. TXD<0> is the least significant bit of lane 0, TXD<8> the least significant bit of lane 1, TXD<16> the least significant bit of lane 2, and TXD<24> the least significant bit of lane 3.

Assertion on a lane of appropriate TXD values when TXC is asserted will cause the PHY to generate code-groups associated with either Idle, Start, Terminate, Sequence, or Error control characters. While the TXC of a lane is de-asserted, TXD of the lane is used to request the PHY to generate code-groups corresponding to the data octet value of TXD. An example of normal frame transmission is illustrated in Figure 46–5.

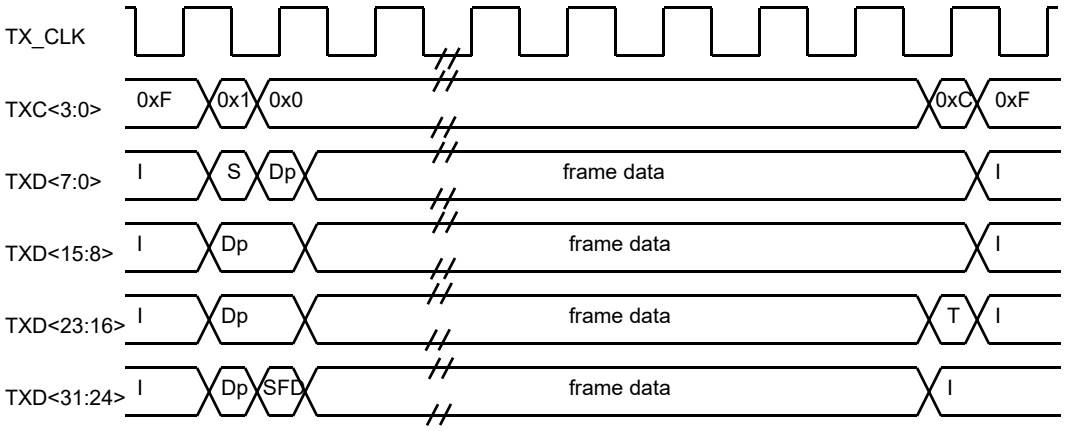
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Table 46–3—Permissible encodings of TXC and TXD

TXC	TXD	Description	PLS_DATA.request parameter
0	00 through FF	Normal data transmission	ZERO, ONE (eight bits)
1	00 through 05	Reserved	—
1	06	Only valid on all four lanes simultaneously to request LPI	No applicable parameter (normal interframe)
1	07	Idle	No applicable parameter (Normal inter-frame)
1	08 through 9B	Reserved	—
1	9C	Sequence (only valid in lane 0)	No applicable parameter (Inter-frame status signal)
1	9D through FA	Reserved	—
1	FB	Start (only valid in lane 0)	No applicable parameter, replaces first eight ZERO, ONE of a frame (preamble octet)
1	FC	Reserved	—
1	FD	Terminate	DATA_COMPLETE
1	FE	Transmit error propagation	No applicable parameter
1	FF	Reserved	—
NOTE—Values in TXD column are in hexadecimal, most significant bit to least significant bit (i.e., <7:0>).			

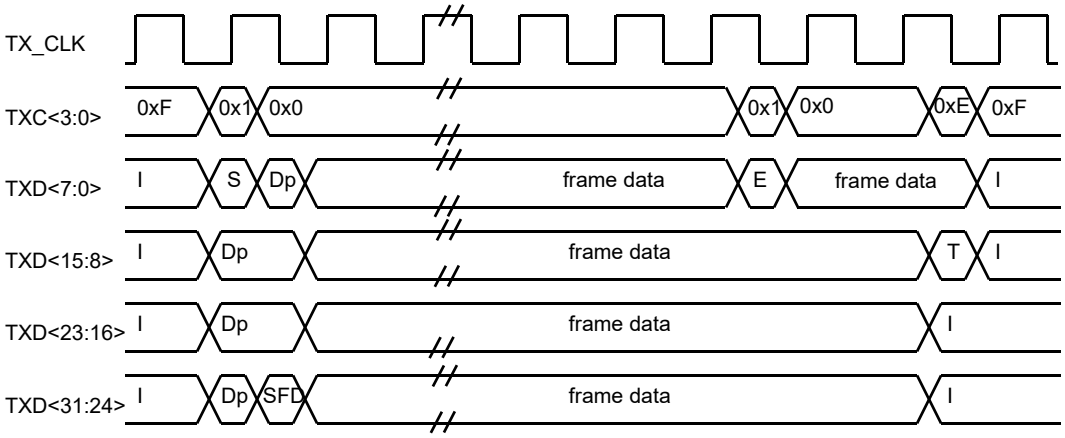
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Figure 46–6 shows the behavior of TXD and TXC during an example transmission of a frame propagating an error.



I: Idle control character, S: Start control character, Dp: preamble Data octet, T: Terminate control character

Figure 46–5—Normal frame transmission



I: Idle control character, S: Start control character, Dp: preamble Data octet, T: Terminate control character, E: Error control character

Figure 46–6—Transmit Error Propagation

46.3.1.4 Start control character alignment

On transmit, it may be necessary for the RS to modify the length of the <inter-frame> in order to align the Start control character (first octet of preamble) on lane 0. This shall be accomplished in one of the following two ways:

1) A MAC implementation may incorporate this RS function into its design and always insert additional idle characters to align the start of preamble on a four byte boundary. Note that this will reduce the effective data rate for certain packet sizes separated with minimum inter-frame spacing.

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2) Alternatively, the RS may maintain the effective data rate by sometimes inserting and sometimes deleting idle characters to align the Start control character. When using this method the RS maintains a Deficit Idle Count (DIC) that represents the cumulative count of idle characters deleted or inserted. The DIC is incremented for each idle character deleted, decremented for each idle character inserted, and the decision of whether to insert or delete idle characters is constrained by bounding the DIC to a minimum value of zero and maximum value of three. Note that this may result in inter-frame spacing observed on the transmit XGMII that is up to three octets shorter than the minimum transmitted inter-frame spacing specified in Clause 4; however, the frequency of shortened inter-frame spacing is constrained by the DIC rules. The DIC is only reset at initialization and is applied regardless of the size of the IPG transmitted by the MAC sublayer. An equivalent technique may be employed to control RS alignment of the Start control character provided that the result is the same as if the RS implemented DIC as described.

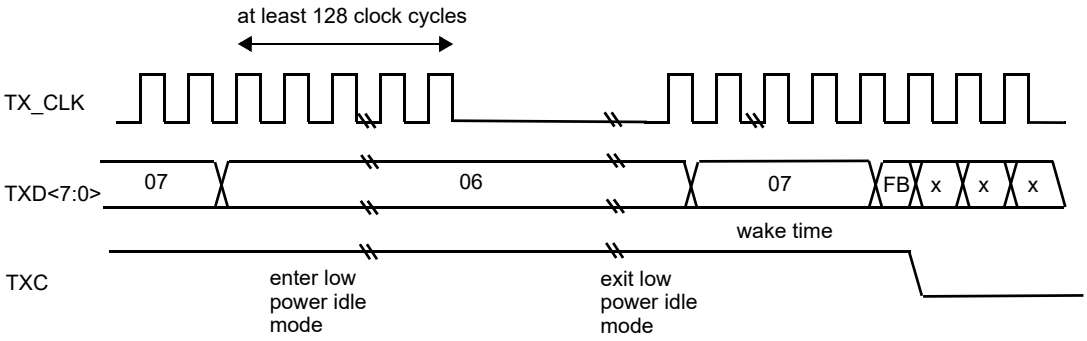
46.3.1.5 Transmit direction LPI transition

LPI operation and the LPI client are described in 78.1. The RS requests the PHY to transition to the LPI state by asserting TXC and setting TXD to 0x06 (in all lanes). The RS maintains the same state for these signals for the entire time that the PHY is to remain in the LPI state.

The RS may halt TX_CLK at any time more than 128 clock cycles after the start of the LPI state as shown in Figure 46–7 if the clock stop capable bit of the attached sublayer is asserted (see 45.2.3.2.1 and 45.2.5.2.1). It is the responsibility of the management entity to ensure that the RS does not halt the TX_CLK if the attached device does not have its stop clock capable bit set. The RS shall restart TX_CLK so that at least one positive transition occurs before it deaserts LPI.

The RS asserts TXC and asserts IDLE on lanes 0–3 in order to make the PHY transition out of the LPI state. The RS should not present a start code for valid transmit data until after the wake-up time specified for the PHY.

Figure 46–7 shows the behavior of TXC and TXD<7:0> during the transition into and out of the LPI state.



NOTE—TXC and TXD are shown for one lane, all four lanes behave identically during LPI.

Figure 46–7—LPI transition

Table 46–3 summarizes the permissible encodings of TXD<31:0>, TXC<3:0>.

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46.3.2 Receive

46.3.2.1 RX_CLK (receive clock)

RX_CLK is a continuous clock that provides the timing reference for the transfer of the RXC<3:0> and RXD<31:0> signals from the PHY to the RS. RXC<3:0> and RXD<31:0> shall be sampled by the RS on both the rising and falling edge of RX_CLK. RX_CLK is sourced by the PHY.

The frequency of RX_CLK may be derived from the received data or it may be that of a nominal clock (e.g., TX_CLK). When the received data rate at the PHY is within tolerance, the RX_CLK frequency shall be $1/64 \times f_{MAC} \pm 100$ ppm, where f_{MAC} is the frequency (in Hz) corresponding to the MAC's nominal bit rate.

There is no need to transition between the recovered clock reference and a nominal clock reference on a frame-by-frame basis. If loss of received signal from the medium causes a PHY to lose the recovered RX_CLK reference, the PHY shall source the RX_CLK from a nominal clock reference. Transitions from nominal clock to recovered clock or from recovered clock to nominal clock shall not decrease the time between adjacent edges of RX_CLK.

NOTE—This standard neither requires nor assumes a guaranteed phase relationship between the RX_CLK and TX_CLK signals. For EEE capability, RX_CLK may be halted according to 46.3.2.4.

46.3.2.2 RXC<3:0> (receive control)

RXC<3:0> indicate that the PHY is presenting either recovered and decoded data or control characters on the XGMII. The RXC signal for a lane shall be de-asserted when a data octet is being received on the corresponding lane and asserted when a control character is being received. In the absence of errors, the RXC signals are de-asserted by the PHY for each octet of the preamble (except the first octet that is replaced with a Start control character) and remain de-asserted while all octets to be received are presented on the lanes of the XGMII. RXC<3:0> are driven by the PHY and shall transition synchronously with respect to both the rising and falling edges of RX_CLK. Table 46–4 specifies the permissible encodings of RXD and RXC for a XGMII receive lane. Additional requirements apply for proper code sequences and in which lanes particular codes are valid (e.g., Start control character is to be aligned to lane 0).

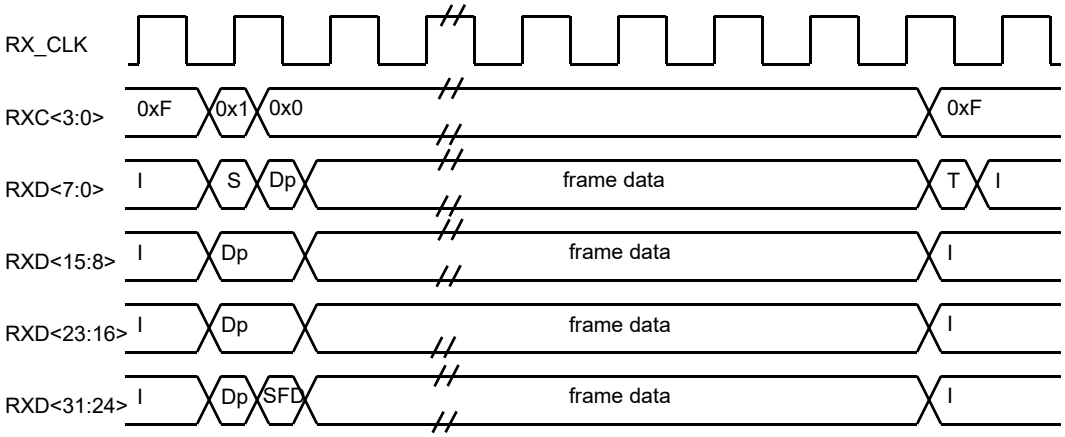
Figure 46–8 shows the behavior of RXC<3:0> during an example frame reception with no errors.

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Table 46–4—Permissible lane encodings of RXD and RXC

RXC	RXD	Description	PLS_DATA.indication parameter
0	00 through FF	Normal data reception	ZERO, ONE (eight bits)
1	00 through 05	Reserved	—
1	06	Only valid on all four lanes simultaneously to indicate LP_IDLE is asserted	No applicable parameter (Normal interframe)
1	07	Idle	No applicable parameter (Normal inter-frame)
1	08 through 9B	Reserved	—
1	9C	Sequence (only valid in lane 0)	No applicable parameter (Inter-frame status signal)
1	9D through FA	Reserved	—
1	FB	Start (only valid in lane 0)	No applicable parameter, first eight ZERO, ONE of a frame (a preamble octet)
1	FC	Reserved	—
1	FD	Terminate	No applicable parameter (Start of inter-frame)
1	FE	Receive error	No applicable parameter
1	FF	Reserved	—

NOTE—Values in RXD column are in hexadecimal, most significant bit to least significant bit (i.e., <7:0>).



I: Idle control character, S: Start control character, Dp: preamble Data octet, T: Terminate control character

Figure 46–8—Basic frame reception

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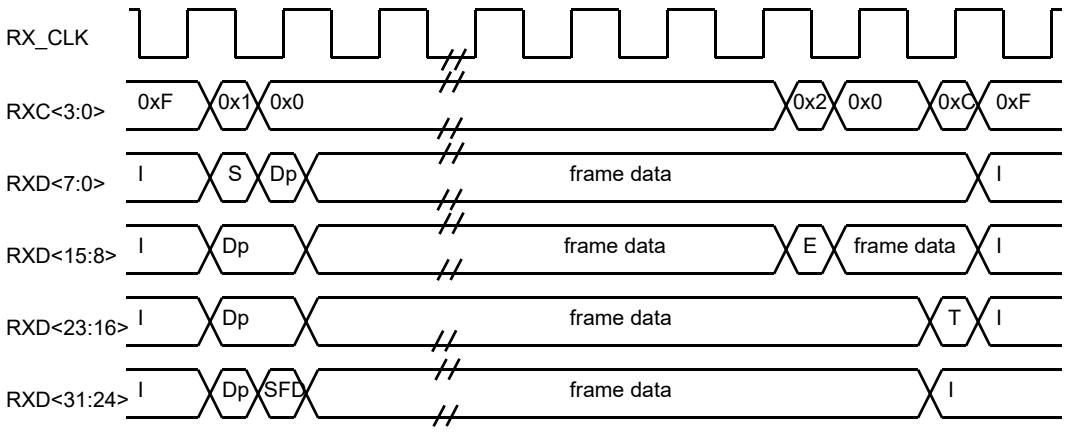
46.3.2.3 RXD (receive data)

RXD is a bundle of 32 data signals (RXD<31:0>) organized into four lanes of eight signals each (RXD<7:0>, RXD<15:8>, RXD<23:16>, and RXD<31:24>) that are driven by the PHY. Each lane is associated with a RXC signal as shown in Table 46–2 and shall be decoded by the RS as shown in Table 46–4. RXD<31:0> shall transition synchronously with respect to both the rising and falling edges of RX_CLK. For each high or low RX_CLK transition, received data and/or control are presented on RXD<31:0> for mapping by the RS. RXD<0> is the least significant bit of lane 0, RXD<8> the least significant bit of lane 1, RXD<16> the least significant bit of lane 2, and RXD<24> the least significant bit of lane 3. Figure 46–8 shows the behavior of RXD<31:0> during frame reception.

While the RXC of a lane is de-asserted, RXD of the lane is used by the RS to generate PLS_DATA.indication transactions. Assertion on a lane of appropriate RXD values when RXC is asserted indicates to the RS the Start control character, Terminate control character, Sequence control character, or Error control character that drive its mapping functions.

RXC of a lane is asserted with the appropriate Error control character encoding on RXD of the lane to indicate an error was detected somewhere in the frame presently being transferred from the PHY to the RS (e.g., a coding error, or any error that the PHY is capable of detecting, and that may otherwise be undetectable at the MAC sublayer).

The effect of an Error control character on the RS is defined in 46.3.3.1. Figure 46–9 shows the behavior of RXC and RXD during the reception of an example frame with an error.



I: Idle control character, S: Start control character, Dp: preamble Data octet, T: Terminate control character, E: Error control character

Figure 46–9—Reception with error

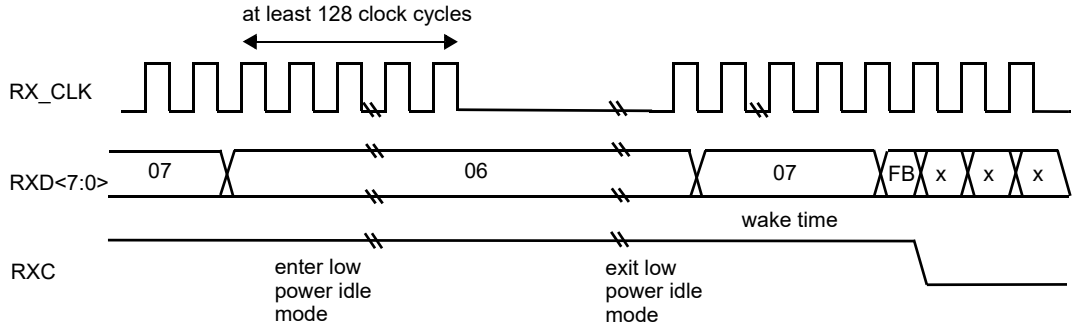
46.3.2.4 Receive direction LPI transition

LPI operation and the LPI client are described in 78.1. When the PHY receives signals from the link partner to indicate transition into the low power state, it indicates this to the RS by asserting RXC and setting RXD to 0x06 (in all lanes). The PHY maintains these signals in this state while it remains in the LPI state. When the PHY receives signals from the link partner to indicate transition out of the LPI state, it indicates this to the RS by asserting RXC and asserting idle on lanes 0–3 to return to a normal interframe state. The RS shall interpret the LPI coding as shown in Table 46–4.

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The PHY or DTE XS may halt RX_CLK at any time more than 128 clock cycles after the start of the LPI state as shown in Figure 46–10 if the clock stop enable bit is asserted (see 45.2.3.1.4 and 45.2.5.1.4). The PHY shall restart RX_CLK so that at least one positive transition occurs before it deasserts LPI.

Figure 46–10 shows the behavior of RXC and RXD<7:0> during LPI transitions.



NOTE 1—RXC and RXD are shown for one lane, all 4 lanes behave identically during LPI.
NOTE 2—In some instances, LPI may be followed by characters other than IDLE during wake time.

Figure 46–10—LPI transition

46.3.3 Error and fault handling

46.3.3.1 Response to error indications by the XGMII

If, during frame reception (i.e., when DATA_VALID_STATUS = DATA_VALID), a control character other than a Terminate control character is signaled on a received lane, the RS shall ensure that the MAC will detect a FrameCheckError in that frame. This requirement may be met by incorporating a function in the RS that produces a received frame data sequence delivered to the MAC sublayer that is guaranteed to not yield a valid CRC result, as specified by the frame check sequence algorithm (see 3.2.8). This data sequence may be produced by substituting data delivered to the MAC. The RS generates eight PLS_DATA.indication primitives for each Error control character received within a frame, and may generate eight PLS_DATA.indication primitives to ensure FrameCheckError when a control character other than Terminate causes the end of the frame.

Other techniques may be employed to respond to a received Error control character provided that the result is that the MAC sublayer behaves as though a FrameCheckError occurred in the received frame.

46.3.3.2 Conditions for generation of transmit Error control characters

If, during the process of transmitting a frame, it is necessary to request that the PHY deliberately corrupt the contents of the frame in such a manner that a receiver will detect the corruption with the highest degree of probability, then an Error control character may be asserted on a transmit lane by the appropriate encoding of the lane's TXD and TXC signals.

46.3.3.3 Response to received invalid frame sequences

The 2.5 Gb/s, 5 Gb/s, or 10 Gb/s PCS adjacent to this RS is required to either preserve the column alignment of the transmitting RS, or align the Start control character to lane 0. The RS shall not indicate DATA_VALID to the MAC for a Start control character received on any other lane. Error free 2.5 Gb/s, 5 Gb/s, or 10 Gb/s operation will not change the SFD alignment in lane 3. A 2.5 Gb/s, 5 Gb/s, or 10 Gb/s MAC/RS

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implementation is not required to process a packet that has an SFD in a position other than lane 3 of the column following the column containing the Start control character.

To support 2.5GBASE-X compatibility with a 1000BASE-X PCS/PMA running 2.5 times faster as described in Annex 127A, a 2.5 Gb/s MAC/RS implementation connected to a 2.5GBASE-X PHY is required to support an SFD received on either lane 2 or lane 3.

46.3.4 Link fault signaling

Link fault signaling operates between the remote RS and the local RS. Faults detected between the remote RS and the local RS are received by the local RS as Local Fault. Only an RS originates Remote Fault signals.

Sublayers within the PHY are capable of detecting faults that render a link unreliable for communication. Upon recognition of a fault condition a PHY sublayer indicates Local Fault status on the data path. When this Local Fault status reaches an RS, the RS stops sending MAC data or LPI, and continuously generates a Remote Fault status on the transmit data path (possibly truncating a MAC frame being transmitted). When Remote Fault or Link Interruption status is received by an RS, the RS stops sending MAC data or LPI, and continuously generates Idle control characters. When the RS no longer receives fault status messages, it returns to normal operation, sending MAC data or LPI.

Status is signaled in a four byte Sequence ordered set as shown in Table 46–5. The PHY indicates Local Fault with a Sequence control character in lane 0 and data characters of 0x00 in lanes 1 and 2 plus a data character of 0x01 in lane 3. The RS indicates a Remote Fault with a Sequence control character in lane 0 and data characters of 0x00 in lanes 1 and 2 plus a data character of 0x02 in lane 3. Though most fault detection is on the receive data path of a PHY, in some specific sublayers, faults can be detected on the transmit side of the PHY. This is also indicated by the PHY with a Local Fault status.

For operation with links that may be temporarily interrupted, optional detection of a third fault condition, Link Interruption, is provided. Link Interruption is indicated by the PHY receive function by continuously sending the Link Interruption ordered set as defined in Table 46–5.

Table 46–5—Sequence ordered sets

Lane 0	Lane 1	Lane 2	Lane 3	Description
Sequence	0x00	0x00	0x00	Reserved
Sequence	0x00	0x00	0x01	Local Fault
Sequence	0x00	0x00	0x02	Remote Fault
Sequence	0x00	0x00	0x03	Link Interruption
Sequence	≥ 0x00	≥ 0x00	≥ 0x04	Reserved
NOTE—Values in Lane 1, Lane 2, and Lane 3 columns are in hexadecimal, most significant bit to least significant bit (i.e., <7:0>). The link fault signaling state diagram allows future standardization of reserved Sequence ordered sets for functions other than link fault indications				

The RS reports the fault status of the link. Local Fault indicates a fault detected on the receive data path between the remote RS and the local RS. Remote Fault indicates a fault on the transmit path between the

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local RS and the remote RS. The RS shall implement the link fault signaling state diagram (see Figure 46–11).

46.3.4.1 Conventions

The notation used in the state diagram follows the conventions of 21.5. The notation ++ after a counter indicates it is to be incremented.

46.3.4.2 Variables and counters

The link fault signaling state diagram uses the following variables and counters:

col_cnt
: A count of the number of columns received not containing a fault_sequence. This counter increments at RX_CLK rate (on both the rising and falling clock transitions) unless reset.

fault_sequence
: A new column received on RXC<3:0> and RXD<31:0> comprising a Sequence ordered set of four bytes and consisting of a Sequence control character in lane 0 and a seq_type in lanes 1, 2, and 3 indicating either Local Faut, Remote Fault, or Link Interruption.

last_seq_type
: The seq_type of the previous Sequence ordered set received
Values: Local Fault; 0x00 in lane 1, 0x00 in lane 2, 0x01 in lane 3.
Remote Fault; 0x00 in lane 1, 0x00 in lane 2, 0x02 in lane 3.
Link Interruption; 0x00 in lane 1, 0x00 in lane 2, 0x03 in lane 3.

link_fault
: An indicator of the fault status.
Values: OK; No fault.
Local Fault; fault detected by the PHY.
Remote Fault; fault detection signaled by the remote RS.
Link Interruption; link temporarily unavailable, signaled by the PHY.

reset
: Condition that is true until such time as the power supply for the device that contains the RS has reached the operating region.
Values: FALSE: The device is completely powered and has not been reset (default).
TRUE: The device has not been completely powered or has been reset.

seq_cnt
: A count of the number of received Sequence ordered sets of the same type.

seq_type
: The value received in the current Sequence ordered set
Values: Local Fault; 0x00 in lane 1, 0x00 in lane 2, 0x01 in lane 3.
Remote Fault; 0x00 in lane 1, 0x00 in lane 2, 0x02 in lane 3.
Link Interruption; 0x00 in lane 1, 0x00 in lane 2, 0x03 in lane 3.

46.3.4.3 State diagram

The link fault signaling state diagram specifies the RS monitoring of RXC<3:0> and RXD<31:0> for Sequence ordered sets. The variable link_fault is set to indicate the value of a received Sequence ordered set when four fault_sequences containing the same fault value have been received with each pair of fault sequences separated by less than 128 columns and no intervening fault_sequences of a different fault value.

The variable link_fault is set to OK following any interval of 128 columns not containing a Remote Fault, Local Fault, or Link Interruption Sequence ordered set.

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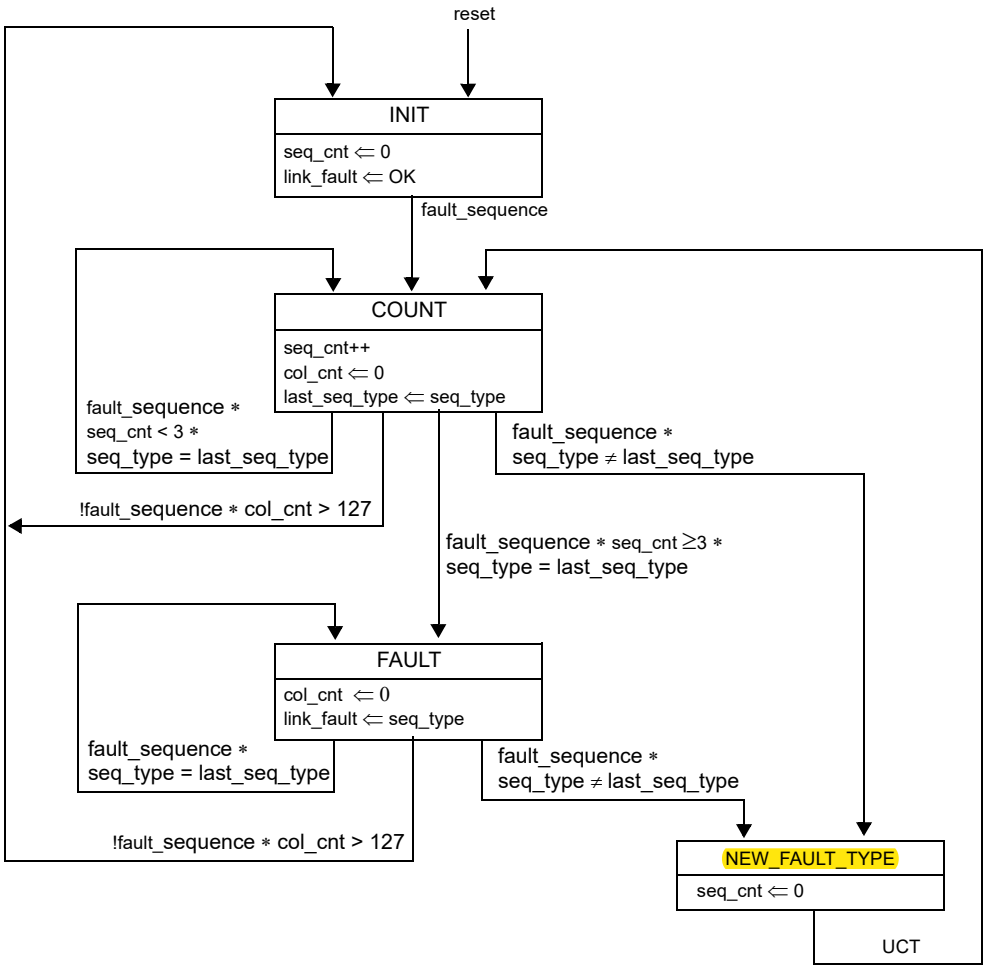


Figure 46–11—Link fault signaling state diagram

The RS output onto TXC<3:0> and TXD<31:0> is controlled by the variable link_fault.

a) link_fault = OK
The RS shall send MAC frames as requested through the PLS service interface. In the absence of MAC frames, the RS shall generate Idle control characters.

b) link_fault = Local Fault
The RS shall continuously generate Remote Fault Sequence ordered sets.

c) link_fault = Remote Fault or link_fault = Link Interruption
The RS shall continuously generate Idle control characters.

46.4 LPI assertion and detection

Certain PHYs support Energy-Efficient Ethernet (see Clause 78). PHYs with EEE capability support LPI assertion and detection. LPI operation and the LPI client are described in 78.1. LPI signaling allows the RS to signal to the PHY and to the link partner that a break in the data stream is expected and components may use this information to enter power-saving modes that require additional time to resume normal operation. Similarly, it allows the LPI client to understand that the link partner has sent such an indication.

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The LPI assertion and detection mechanism fits conceptually between the PLS Service Primitives and the XGMII signals as shown in Figure 46–12.

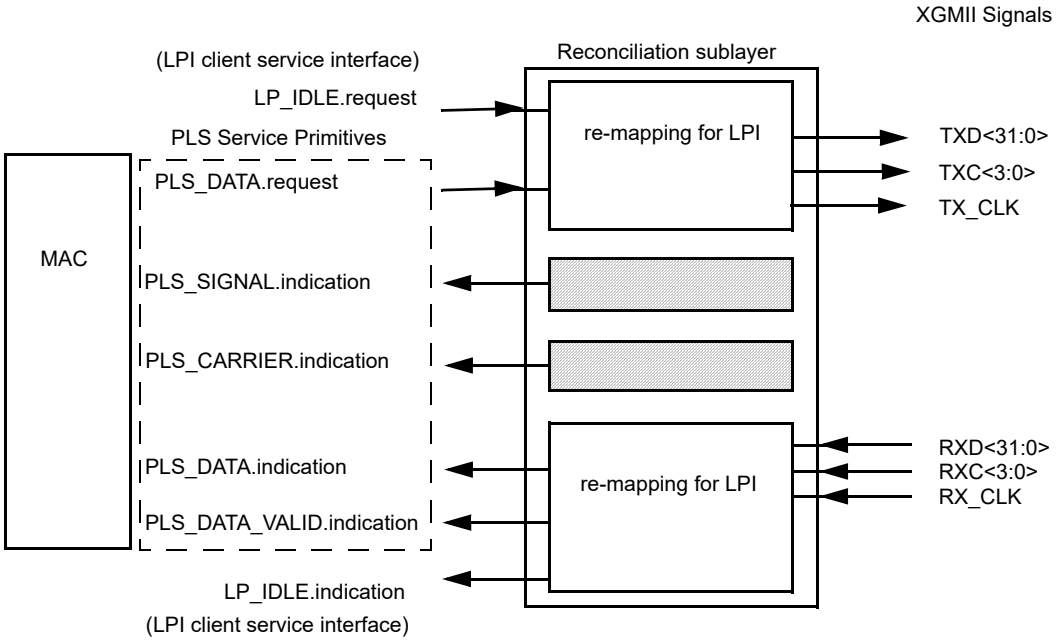


Figure 46–12—LPI assertion and detection mechanism

The definition of TXC<3:0> and TXD<31:0> is derived from the state of PLS_DATA.request (46.1.7), except when it is overridden by an assertion of LP_IDLE.request.

Similarly, RXC<3:0> and RXD<31:0> are mapped to PLS_DATA.indication except when LP_IDLE is detected

PLS_CARRIER.indication(CARRIER_STATUS) will be set to CARRIER_ON when the link is in LPI mode. See 46.1.7.3.

The timing of PLS_CARRIER.indication when used for the LPI function is controlled by the LPI transmit state diagram.

46.4.1 LPI messages

LP_IDLE.indication(LPI_INDICATION)

A primitive that indicates to the LPI client that the PHY has detected the assertion or de-assertion of LPI from the link partner.

Values: DEASSERT: The link partner is operating with normal interframe behavior (default).
ASSERT: The link partner has asserted LPI.

LP_IDLE.request(LPI_REQUEST)

The LPI_REQUEST parameter can take one of two values: ASSERT or DE-ASSERT. ASSERT initiates the signaling of LPI to the link partner. DE-ASSERT stops the signaling of LPI to the link partner. The effect of receipt of this primitive is undefined if link_status is not OK (see 28.2.6.1.1) or within 1 s of the change of link_status to OK.

46.4.2 Transmit LPI state diagram

The operation of LPI in the PHY requires that the MAC does not send valid data for a time after LPI has been de-asserted as governed by resolved Transmit T_{w_sys} defined in 78.4.2.3.

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This wake-up time is enforced by the transmit LPI state diagram using CARRIER_SENSE.indication. The implementation shall conform to the behavior described by the transmit LPI state diagram shown in Figure 46–13.

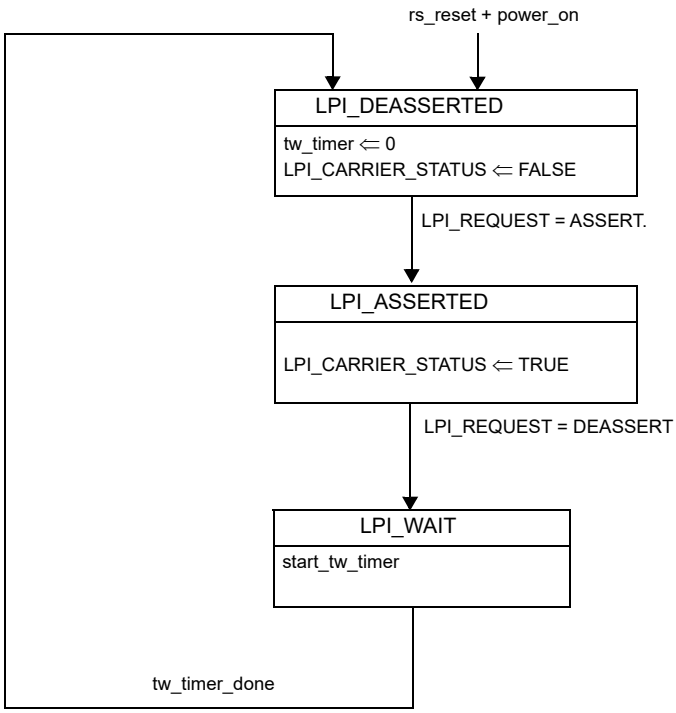


Figure 46–13—Transmit LPI state diagram

46.4.2.1 Variables and counters

The transmit LPI state diagram uses the following variables and counters:

LPI_CARRIER_STATUS
: The LPI_CARRIER_STATUS variable indicates how the CARRIER_STATUS parameter is controlled by the LPI_REQUEST parameter. The LPI_CARRIER_STATUS is either TRUE or FALSE as determined by the Transmit LPI state diagram in Figure 46–13.

power_on
: Condition that is true until such time as the power supply for the device that contains the RS has reached the operating region.
Values: FALSE: The device is completely powered (default).
TRUE: The device has not been completely powered.

rs_reset
: Used by management to control the resetting of the RS.
Values: FALSE: Do not reset the RS (default).
TRUE: Reset the RS.

tw_timer
: A timer that counts, in microseconds, the time since the de-assertion of LPI. The terminal count of the timer is the value of the resolved $T_{w_sys_tx}$ as defined in 78.2. If DTE XS XAUI stop enable bit is asserted (5.0.9), the terminal count of the timer is the value of the resolved $T_{w_sys_tx}$ as defined in 78.2 plus additional time equal to $T_{w_sys_tx} - T_{w_sys_rx}$ for the XGXS as shown in Table 78–4.

The signal tw_timer_done is asserted when tw_timer reaches its terminal count.

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46.4.3 Considerations for transmit system behavior

The transmit system should expect that egress data flow will be halted for at least resolved $T_{w_sys_tx}$ (see 78.2) time, in microseconds, after it requests the de-assertion of LPI. Buffering and queue management should be designed to accommodate this.

46.4.4 Considerations for receive system behavior

The mapping function of the Reconciliation Sublayer shall continue to signal IDLE on PLS_DATA.indicate while it is detecting LP_IDLE on the XGMII. The receive system should be aware that data frames may arrive at the XGMII following the de-assertion of LPI_INDICATION with a delay corresponding to the link partner's resolved $T_{w_sys_rx}$ (as specified in 78.5) time, in microseconds.

If the PHY XS XAUI stop enable bit (4.0.9) is asserted, the PHY XS may stop signaling on the XAUI in the receive direction to conserve energy. The receiver should negotiate an additional 9.5 μs for the remote T_{w_sys} (equal to $T_{w_sys_tx} - T_{w_sys_rx}$ for the XGXS as shown in Table 78–4) before setting the PHY XS XAUI stop enable bit.

46.5 XGMII electrical characteristics

The electrical characteristics of the XGMII are specified such that the XGMII can be applied within a variety of 2.5 Gb/s, 5 Gb/s, and 10 Gb/s equipment types. The electrical specifications are selected for an integrated circuit to integrated circuit application. The electrical characteristics specified in this clause apply to all XGMII signals.

When implemented as a chip-to-chip interface, the XGMII uses High Speed Transceiver Logic (HSTL), specified for a 1.5 volt output buffer supply voltage. XGMII chip-to-chip signals shall comply with EIA/JEDEC Standard EIA/JESD8-6 using Class I, output buffers. Output impedance shall be greater than 38Ω to assure acceptable overshoot and undershoot performance in an unterminated interconnection.

The thresholds and parametric values for HSTL XGMII signals are shown in the illustrative Figure 46–14 and example Table 46–6.

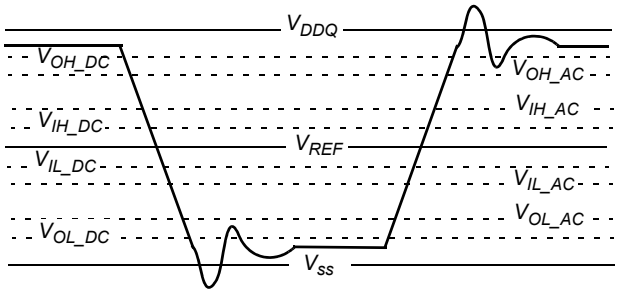


Figure 46–14—Electrical characteristics (illustrative)

The HSTL specification shows a number of termination options for different applications. Figure 46–15 illustrates a possible XGMII circuit topology. Unterminated interconnection is recommended.

The XGMII chip-to-chip signals shall meet the timing requirements shown in Figure 46–16. All XGMII timing measurements shall be made at the XGMII driver output with the optional termination shown in Figure 46–15 and with a capacitive load from all sources of 10pF and are specified relative to the $V_{IL_AC(max)}$ and $V_{IH_AC(min)}$ thresholds.

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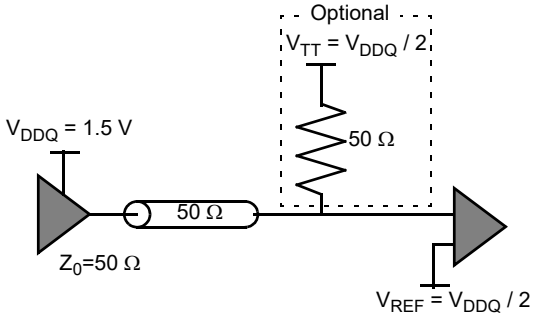
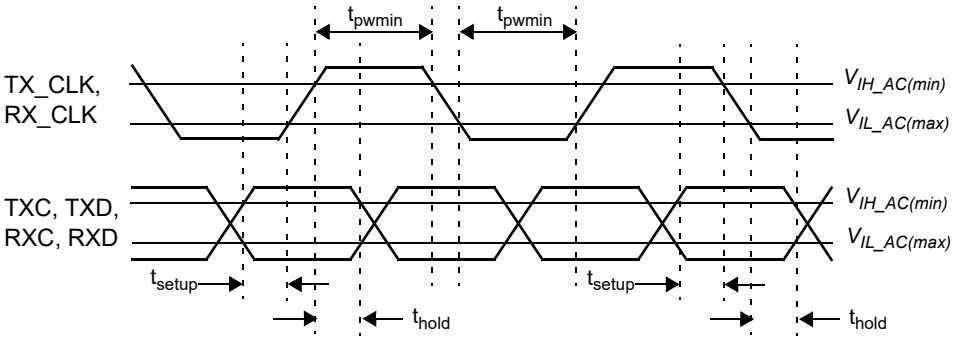


Figure 46–15—Circuit topology example



Symbol	Driver	Receiver	Units
t_{setup}	960	480	ps
t_{hold}	960	480	ps
t_{pwmin}	2.5	—	ns

Figure 46–16—TX_CLK and RX_CLK timing parameters

Table 46–6—Example DC and AC specifications

Symbol	Parameter	Min	Nom	Max	Units
V_{DDQ}	Output supply voltage	1.4	1.5	1.6	V
V_{REF}	Input reference voltage	0.68	0.75	0.90	V
V_{IH_DC}	DC input logic high	V_{REF}+0.10	—	V_{DDQ}+0.3	V
V_{IL_DC}	DC input logic low	–0.30	—	V_{REF}–0.1	V
V_{IH_AC}	AC input logic high	V_{REF}+0.20	—	—	V
V_{IL_AC}	AC input logic low	—	—	V_{REF}–0.20	V

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46.6 Protocol implementation conformance statement (PICS) proforma for Clause 46, Reconciliation Sublayer (RS) and 10 Gigabit Media Independent Interface (XGMII)[88]

46.6.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 46, Reconciliation Sublayer (RS) and 10 Gigabit Media Independent Interface (XGMII), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

46.6.2 Identification

46.6.2.1 Implementation identification

Supplier[1]	
Contact point for inquiries about the PICS[1]	
Implementation Name(s) and Version(s)[1,3]	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s)[2]	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

46.6.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 46, Reconciliation Sublayer (RS) and 10 Gigabit Media Independent Interface (XGMII)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [ ] Yes [ ] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	

Date of Statement	

[88] *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

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46.6.2.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*PHY	PHY support of XGMII	46.2, 46.3		O	Yes [ ] No [ ]
*RS	Reconciliation Sublayer support of XGMII	46.2, 46.3		O	Yes [ ] No [ ]
*XGE	XGMII electrical interface	46.4		O	Yes [ ] No [ ]
*LPI	Implementation of LPI	46.1.7		O	Yes [ ] No [ ]

46.6.3 PICS proforma tables for Reconciliation Sublayer and 10 Gigabit Media Independent Interface

46.6.3.1 General

Item	Feature	Subclause	Value/Comment	Status	Support
G1	PHY support of 10 Gb/s MAC data rate	46.1.3	Support MAC data rate of 10 Gb/s	PHY:O.1	Yes [ ] N/A [ ]
G2	PHY support of 5 Gb/s MAC data rate	46.1.3	Support MAC data rate of 5 Gb/s	PHY:O.1	Yes [ ] N/A [ ]
G3	PHY support of 2.5 Gb/s MAC data rate	46.1.3	Support MAC data rate of 2.5 Gb/s	PHY:O.1	Yes [ ] N/A [ ]
G4	Cumulative MAC Control, MAC and RS round-trip delay	46.1.4	Per Table 46–1	RS:M	Yes [ ] N/A [ ]
G5	Lane structure	46.1.6	Per Table 46–2	M	Yes [ ]

46.6.3.2 Mapping of PLS service primitives

Item	Feature	Subclause	Value/Comment	Status	Support
PL1	Mapping to Clause 6	46.1.7	RS implements mapping to Clause 6 PLS service primitives	RS:M	Yes [ ] N/A [ ]
PL2	Mapping of PLS_DATA.requests	46.1.7.1.4	In sequence TXD<0> to TXD<31>	RS:M	Yes [ ] N/A [ ]
PL3	Start control character creation	46.1.7.1.4	First octet of preamble converted to Start control character	RS:M	Yes [ ] N/A [ ]
PL4	TXD and TXC generation	46.1.7.1.4	For each 32 PLS_DATA.requests	RS:M	Yes [ ] N/A [ ]

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Item	Feature	Subclause	Value/Comment	Status	Support
PL5	Terminate control character creation	46.1.7.1.4	DATA_COMPLETE causes creation of Terminate control character in next lane in sequence	RS:M	Yes [ ] N/A [ ]
PL6	Mapping RXD to PLS_DATA.incicates	46.1.7.2.3	Create PLS_DATA.increments in sequence from RXD<0> to RXD<31>	RS:M	Yes [ ] N/A [ ]
PL7	PLS_DATA.indication generation	46.1.7.2.3	Generate 32 PLS_DATA.indications for each RXD<0:31> until Terminate then generating 0, 8, 16, or 24	RS:M	Yes [ ] N/A [ ]
PL8	Start control character conversion	46.1.7.2.3	Convert valid Start control character to preamble before generating PLS_DATA.indications	RS:M	Yes [ ] N/A [ ]
PL9	Terminate control character	46.1.7.2.3	No PLS_DATA.indications generated	RS:M	Yes [ ] N/A [ ]
PL10	PLS_DATA_VALID.indicatio n generation	46.1.7.5.3	On change of value of DATA_VALID_STATUS	RS:M	Yes [ ] N/A [ ]
PL11	DATA_VALID_STATUS	46.1.7.5.3	Value of DATA_VALID on a lane 0 Start control character preceded by four idles or a Sequence ordered set	RS:M	Yes [ ] N/A [ ]
PL12	DATA_VALID_STATUS	46.1.7.5.3	Value of DATA_NOT_VALID on any control character but Error	RS:M	Yes [ ] N/A [ ]
PL13	Frame not ending with Terminate control character	46.1.7.5.3	Ensure MAC detects CRC error	RS:M	Yes [ ] N/A [ ]

46.6.3.3 Data stream structure

Item	Feature	Subclause	Value/Comment	Status	Support
DS1	Frame transfer	46.2	Within XGMII data stream	RS:M	Yes [ ] N/A [ ]
DS2	Bit mapping	46.2	Per Figure 46–4	RS:M	Yes [ ] N/A [ ]
DS3	Content of <data>	46.2.3	Consist of data octets	RS:M	Yes [ ] N/A [ ]
DS4	Recognition of <efd>	46.2.4	Terminate recognized in any lane	RS:M	Yes [ ] N/A [ ]

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46.6.3.4 LPI functions

Item	Feature	Subclause	Value/Comment	Status	Support
L1	Assertion of LPI in Tx direction	46.3.1.2	As defined in Table 46–3	LPI:M	Yes [ ] N/A [ ]
L2	Assertion of LPI in Rx direction	46.3.2.2	As defined in Table 46–4	LPI:M	Yes [ ] N/A [ ]
*L3	TX_CLK stoppable during LPI	46.3.1.5	At least 128 cycles after LPI assertion	LPI:O	Yes [ ] No [ ]
L4	TX_CLK restart before LPI deassert	46.3.1.5	At least 1 positive edge before LPI deassertion	L3:M	Yes [ ] N/A [ ]
L5	RX_CLK stoppable during LPI	46.3.2.4		LPI:O	Yes [ ] No [ ]

46.6.3.5 Link Interruption

Item	Feature	Subclause	Value/Comment	Status	Support
LINT	Detection of Link Interruption	46.3.4		O	Yes [ ] No [ ]
LINT1	CARRIER_STATUS response to Link Interruption	46.1.7.3	Set to CARRIER_ON if link_fault is Link Interruption	LINT:O	Yes [ ] No [ ]

46.6.3.6 XGMII signal functional specifications

NOTE—An XGXS adjacent to an RS exhibits the characteristics of a PHY for the items in this subclause, and an XGXS adjacent to a PCS exhibits the characteristics of an RS for the items in this subclause.

Item	Feature	Subclause	Value/Comment	Status	Support
FS1	TX_CLK active edges	46.3.1.1	TXD and TXC sampled on both edges of TX_CLK	XGE:M	Yes [ ] N/A [ ]
FS2	TX_CLK frequency	46.3.1.1	$1/64 \times f_{MAC} \pm 100$ ppm, where f_{MAC} is the frequency (in Hz) corresponding to the MAC's nominal bit rate	XGE:M	Yes [ ] N/A [ ]
FS3	TXC assertion and de-assertion	46.3.1.2	De-asserted for data, asserted for control character	RS:M	Yes [ ] N/A [ ]
FS4	TXC clock	46.3.1.2	Synchronous to TX_CLK	XGE:M	Yes [ ] N/A [ ]
FS5	TXD encoding	46.3.1.3	Per Table 46–3	RS:M	Yes [ ] N/A [ ]
FS6	TXD clock	46.3.1.3	Synchronous to TX_CLK	XGE:M	Yes [ ] N/A [ ]

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Item	Feature	Subclause	Value/Comment	Status	Support
FS7	Start alignment	46.3.1.4	Start control character aligned to lane 0	RS:M	Yes [ ] N/A [ ]
FS8	RX_CLK active edges	46.3.2.1	RXD and RXC sampled on both edges of RX_CLK	XGE:M	Yes [ ] N/A [ ]
FS9	RX_CLK frequency	46.3.2.1	$1/64 \times f_{MAC} \pm 100$ ppm, where f_{MAC} is the frequency (in Hz) corresponding to the MAC's nominal bit rate when received data rate is within tolerance	XGE:M	Yes [ ] N/A [ ]
FS10	Loss of receive signal	46.3.2.1	Source RX_CLK from nominal clock	PHY:M	Yes [ ] N/A [ ]
FS11	Transition between clock sources	46.3.2.1	No decrease of RX_CLK period when switching sources	PHY:M	Yes [ ] N/A [ ]
FS12	RXC assertion and de-assertion	46.3.2.2	De-asserted for data, asserted for control character	PHY:M	Yes [ ] N/A [ ]
FS13	RXC clock	46.3.2.2	Synchronous to RX_CLK	XGE:M	Yes [ ] N/A [ ]
FS14	RXD decoding	46.3.2.3	Per Table 46–4	RS:M	Yes [ ] N/A [ ]
FS15	RXD clock	46.3.2.3	Synchronous to RX_CLK	XGE:M	Yes [ ] N/A [ ]
FS16	Received Error control character	46.3.3.1	RS cause MAC FrameCheckError	RS:M	Yes [ ] N/A [ ]
FS17	DATA_VALID assertion	46.3.3.3	RS not assert DATA_VALID unless Start control character in lane 0	RS:M	Yes [ ] N/A [ ]

46.6.3.7 Link fault signaling state diagram

Item	Feature	Subclause	Value/Comment	Status	Support
LF1	Link fault signaling state diagram	46.3.4	Implement per Figure 46–11	RS:M	Yes [ ] N/A [ ]
LF2	link_fault = OK and MAC frames	46.3.4.3	RS services MAC frame transmission requests	RS:M	Yes [ ] N/A [ ]
LF3	link_fault = OK and no MAC frames	46.3.4.3	In absence of MAC frames, RS transmits Idle control characters	RS:M	Yes [ ] N/A [ ]
LF4	link_fault = Local Fault	46.3.4.3	RS transmits continuous Remote Fault Sequence ordered sets	RS:M	Yes [ ] N/A [ ]
LF5	link_fault = Remote Fault	46.3.4.3	RS transmits continuous Idle control characters	RS:M	Yes [ ] N/A [ ]

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46.6.3.8 Electrical characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
EC1	Referenced standard for signals	46.5	Signals to be compliant with EIA/JESD8-6 using Class I, output buffers	XGE:M	Yes [ ] N/A [ ]
EC2	Output impedance	46.5	Signal output impedance of at least 38 Ω	XGE:M	Yes [ ] N/A [ ]
EC3	Signal timing	46.5	Per Figure 46–16	XGE:M	Yes [ ] N/A [ ]
EC4	Signal measurement	46.5	At driver output with optional termination per Figure 46–15 and with a capacitive load from all sources of 10 pF	XGE:M	Yes [ ] N/A [ ]

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